

Customer Shopping Behavior Analysis

A complete data analytics report using Python, SQL, and the supplied Customer Behavior HTML Dashboard.

3,900	18	\$59.76	3.75 / 5.0
Total Records	Variables	Avg. Purchase	Avg. Rating

1. Project Overview

This project analyses customer shopping behaviour using transactional data from 3,900 purchases across product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behaviour to guide strategic business decisions.

Business Problem

- Identify key trends in purchasing patterns across demographics.
- Determine which customer segments generate the highest revenue.
- Evaluate the effectiveness of subscription programmes and discount policies.
- Improve customer engagement and optimise marketing or product strategies.

Analytical Approach

- Data collection and loading
- Exploratory data analysis and cleaning with Python/Pandas
- Database ingestion into PostgreSQL
- Advanced SQL analytics
- Interactive dashboard review using the supplied HTML dashboard
- Strategic business recommendations

2. Dataset Summary

Property	Detail
Source file	customer_shopping_behavior.csv
Total rows	3,900
Total columns	18
Missing data	37 null values in review_rating
Data types	Numeric, categorical, and text

Key Feature Groups

Customer demographics: age, gender, location, and subscription status.

Purchase details: item purchased, category, purchase amount, season, size, and color.

Shopping behaviour: discount applied, previous purchases, frequency of purchases, review rating, and shipping type.

Summary Statistics

Stat	Customer ID	Age	Purchase Amt	Review Rating	Previous Purchases
count	3,900	3,900	3,900	3,863	3,900
mean	-	44.07	\$59.76	3.75	-
std	-	15.21	23.69	0.72	-
min	1	18	\$20.00	2.5	1
median	1,950	44	\$60.00	3.8	25
max	3,900	70	\$100.00	5.0	50

3. Exploratory Data Analysis using Python

Data preparation and cleaning was performed in Python using Pandas. Initial checks confirmed 18 columns, 3,900 rows, 37 missing review_rating values, and no duplicate rows.

Method	Purpose	Result
df.info()	Column data types and nulls	18 columns, 3,900 rows, 37 nulls in review_rating
df.describe()	Summary statistics	Mean purchase \$59.76, average rating 3.75
df.shape	Dimensions	(3900, 18)
df.duplicated().sum()	Duplicate check	0 duplicates found

Cleaning and Feature Engineering

Missing review ratings were imputed using category-level medians to preserve product-category rating patterns.

Column names were standardized to lowercase snake_case for consistency across Python and SQL workflows.

Customers were grouped into Young Adult, Middle-aged, Adult, and Senior age bands.

Text-based purchase frequency values were mapped to numeric day intervals.

promo_code_used was dropped because it mirrored discount_applied with 100% redundancy.

Age Group Feature

Age Group	Age Range	Customer Count	Revenue Share
Young Adult	Under 25	486	12.6%
Middle-aged	25-39	1,116	28.8%
Adult	40-59	1,510	38.5%
Senior	60+	788	20.1%

4. Advanced Data Analysis using SQL

The cleaned data was ingested into PostgreSQL using SQLAlchemy and psycopg2. SQL queries were used to compare revenue by gender, shipping method, subscription status, discount dependency, loyalty segment, product category, and age group.

Revenue Split by Gender

Gender	Customer Count	Total Revenue	Revenue Share	Avg. Spend
Male	2,652	\$157,890	67.7%	\$59.54
Female	1,248	\$75,191	32.3%	\$60.25

Insight: the gender revenue gap is volume-driven. Average spend is nearly identical, with female customers slightly higher per visit.

Shipping Method ROI Comparison

Shipping Type	Avg. Purchase	vs. Standard	Assessment
2-Day Shipping	\$60.73	+\$2.27	Highest AOV
Express	\$60.48	+\$2.02	Strong
Free Shipping	\$60.41	+\$1.95	Strong
Store Pickup	\$59.89	+\$1.43	Solid
Next Day Air	\$58.63	+\$0.17	Near baseline
Standard	\$58.46	Baseline	Baseline

Subscription vs. Direct Purchase Revenue

Subscription	Customers	Avg. Spend	Total Revenue	Revenue Share
Non-subscribed	2,847	\$59.87	\$170,436	73.1%
Subscribed	1,053	\$59.49	\$62,645	26.9%

Critical finding: subscribers spend \$0.38 less per transaction than non-subscribers, so the programme needs a stronger value proposition.

Top Promotion-Dependent Products

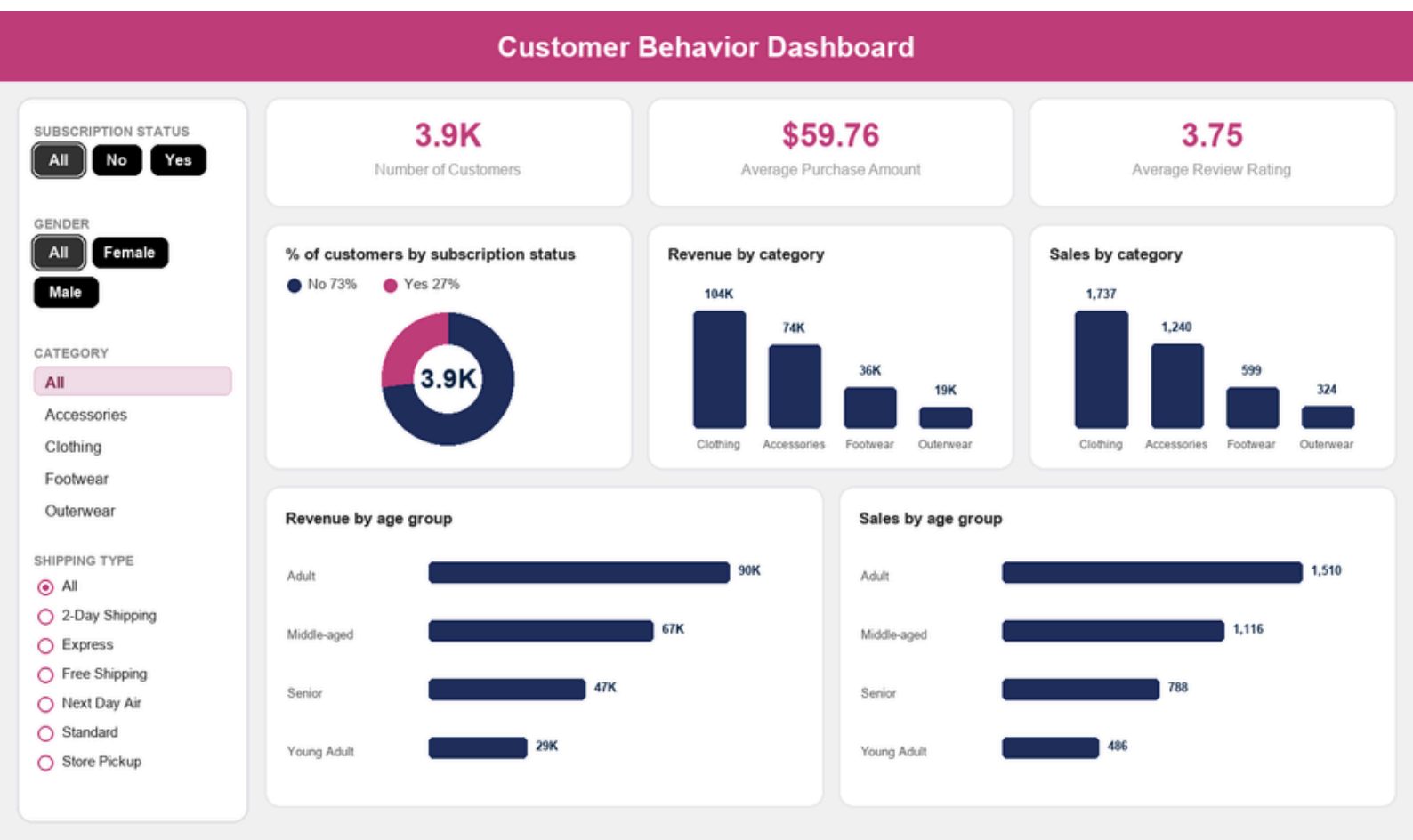
Rank	Item	Discount Rate	Risk Level
1	Hat	50.0%	Critical
2	Sneakers	49.7%	Critical
3	Coat	49.1%	High
4	Sweater	48.2%	High
5	Pants	47.4%	Medium-high

Margin alert: systematic discounting near 50% can train customers to wait for promotions.

Customer Loyalty Segmentation

Segment	Qualification	Customers	Share	Implication
Loyal	11+ previous purchases	3,116	79.9%	Core revenue base
Returning	2-10 previous purchases	701	18.0%	Growth opportunity
New	First purchase only	83	2.1%	Acquisition risk

5. Interactive Visualisations and KPIs



Dashboard inserted from the supplied customer_behavior_dashboard_v4.html content. The visual shows the dashboard slicers, KPI cards, subscription split, category performance, and age-group performance.

Dashboard Headline KPIs

KPI	Value	Interpretation
Number of customers	3.9K	Total records in the shopping dataset
Average purchase amount	\$59.76	Mean transaction value
Average review rating	3.75	Customer satisfaction out of 5.0
Subscription split	No 73% / Yes 27%	Most customers are not subscribed
Clothing revenue share	44.7%	Largest revenue category

Category Revenue Breakdown

Category	Revenue	Revenue Share	Sales	Avg. Sale Value
Clothing	\$104,264	44.7%	1,737	\$60.03
Accessories	\$74,200	31.8%	1,240	\$59.84
Footwear	\$36,093	15.5%	599	\$60.26
Outerwear	\$18,524	7.9%	324	\$57.17

6. Business Recommendations

- Boost subscriptions:** Promote exclusive benefits such as free Express shipping, cashback on repeat purchases, or early access to new arrivals.
- Customer loyalty programmes:** Use milestone rewards to move Returning customers into the Loyal segment and protect long-term revenue.
- Review discount policy:** Run pricing tests for Hat, Sneakers, Coat, Sweater, and Pants to reduce near-50% discount dependency.
- Product positioning:** Use top category leaders such as Blouse, Jewelry, Sandals, and Jacket in seasonal campaigns.
- Targeted marketing:** Prioritise Adults and Middle-aged customers by volume, and use premium shipping behavior for upsell campaigns.

Recommendation Priority Matrix

Recommendation	Expected Impact	Effort	Priority
Revamp subscription benefits	High: unlocks non-sub revenue	Medium	P1
Reprice discount-dependent SKUs	High: margin recovery	Low	P1
Loyalty programme	Medium: LTV increase	Medium	P2
New customer acquisition	High: closes acquisition gap	High	P2
Targeted age/shipping marketing	Medium: AOV lift	Low	P3
Outerwear portfolio expansion	Medium: new revenue stream	High	P3

Conclusion

This report preserves the original customer shopping behavior analysis content while replacing the dashboard section with the supplied HTML dashboard visual. The main strategic priorities remain subscription redesign, discount repricing, new customer acquisition, and targeted marketing based on category, age-group, and shipping behavior.

Revenue: \$233,081 | Customers: 3,900 | Loyalty: 79.9% | Satisfaction: 3.75 / 5.0

Sources: Customer_Shopping_Behavior_Analysis.docx and customer_behavior_dashboard_v4.html. Report generated May 20, 2026.